

Changes that were built, measured, and declined

Six candidate improvements were implemented and **measured** — not sketched — each in its own git worktree. Each was then attacked by three independent reviewers asked to check that the numbers came from a real run, to hunt for benchmark-tuning or copied code, and to map what would break among the gates currently passing.

Eighteen reviewers returned **6 tuning votes, 9 code-copying votes and 7 took-from-truth votes**. Two candidates came back clean on all four flags and shipped (commit [b40bd3f](#)). Four did not, and this page is why — because a change that moves a gate and should not ship is more interesting than one that does.

One of the four was later **split**: §2's principled half was rewritten independently and shipped ([9078b86](#)), while the half that actually flipped its gate stayed declined. That is the shape this page is really about — not "reject the change" but "separate the part that is correct from the part that is only favourable".

Every number below was measured in a worktree. They are available to a reader who wants to check them: the commits are not in this history, but the method and the measured outcome are.

1. Adaptive keyframe selection — DECLINED for tuning

What it was. Replace the fixed frame stride (keep every 3rd frame) with adaptive selection: keep a frame when the pose has moved or rotated enough since the last kept one. This is the right idea. Our fixed stride has a measured failure: a capture whose camera sweep is periodic can alias with the sampling, and `scanplan run` exits with "no floor found" on a capture that plainly shows the floor. `scanplan/synthetic.py` documents that at length.

What it measured. Genuinely good. A-RUNTIME improved from 52.7 s to 40.8 s — real headroom against a 60 s budget. The aliasing case was fixed: where stride 3 raises `CaptureError`, adaptive selection measures 12.015 m² against 12.00 m² of truth. No currently-passing gate regressed. One reviewer independently re-derived every keyframe count from the odometry and matched the report exactly.

Why it is declined anyway. The implementation's own sensitivity sweep is the exhibit. Five configurations were tried. The shipped threshold, 2.9°, is **the unique point in the entire swept space that clears A-RUNTIME, G-REPEAT-FOOTPRINT and G-REPEAT-ROOMS simultaneously**. The threshold the stated derivation actually produces — 2σ , or $\text{atan}(0.05) = 2.862^\circ$ — reads 5 rooms against 4 and leaves G-REPEAT-ROOMS **NOT MET**. The tighter 1σ option clears both repeatability gates but costs 111.4 s on one capture, blowing the runtime budget.

So the derivation was presented as the justification while the value was selected by its effect on three gates. All three reviewers called it tuning. That is the definition: a constant whose only distinguishing property is that it passes.

The honest position. Adaptive selection is better than a fixed stride and we should have it. It cannot be adopted with a threshold chosen this way. Adopting it with the derived 2.862° costs a currently-passing gate, which is the real price and would have to be paid openly. Neither was possible in the time left, so the fixed stride stays, with the `--stride` escape hatch and the diagnostic that names stride aliasing when it happens.

2. Normals-based wall-plane selection — SHIPPED, in two stages, the second with its definition stated

What it was. A-WALL-LIDAR reads 7/12. Our plane finder takes the two densest 1 cm histogram peaks, which is a raw argmax: where two parallel surfaces 0.19–0.99 m apart have near-equal support, the choice turns on a margin as small as 2.2%, and the device and laser clouds can land differently. The fix adds per-point normals, a facing test and extent gates — the approach `cozmo-scan` uses, which reports 4/4 on this gate.

What it measured. The principled part works: worst row 975 mm → 612 mm, median `|error|` 16.4 mm → **6.0 mm**. Tying the non-max suppression radius to the estimator's own window (4 bins instead of a free 10) is a strict improvement and independently defensible.

Why it is declined. Two reasons, and one reviewer rejected it outright.

The gate only reaches MET (11/11) via a *second* change: let the device choose which two surfaces to measure and have the laser measure **those same two**. That is defensible as a definition — A-WALL-LIDAR is a length gate, not a selection-agreement gate — but it *relaxes what the gate penalises*, and it was introduced by the same author who then reported the gate as passing. The denominator also fell from 12 to 11. Three reviewers called it tuning.

Separately, the provenance claim was false. The normal computation in the new module is a line-for-line transliteration of `cozmo-scan's fusion.py:frame_points` — identical slicing, the same epsilon, the same 0.04 depth-step rejection, the same sign flip — and four of ten lines are byte-identical. It was labelled "reimplemented approach". Computing normals from a depth grid by central differences is a standard method with few degrees of freedom, so the overlap is partly unavoidable; the mislabelling is not.

What happened next. The principled half was then written independently and **shipped** (`bench/wall_normals.py`): normals + facing + extent gates, suppression radius tied to the estimator's window, with each cloud still selecting its own wall pair. Measured:

	before	after
within gate	7/12	8/10
median error	16.4 mm	9.8 mm
worst row	975.4 mm	225.3 mm
where both clouds chose the same walls	6/9, 7.5 mm	8/8, 8.1 mm

The 975 mm row is now +3.4 mm.

And then the second half was shipped too, with the narrowing stated. On review of this page, the judgement was overruled deliberately: the shared-selection definition is defensible on its own terms — A-WALL-LIDAR is a *length* gate, and whether we pick the right pair of walls is a separate property that is measured separately. What made it unacceptable the first time was not the definition but the packaging: it arrived bundled with a false provenance claim and was reported as a plain pass.

It is now reported as **10/10 MET under the shared-selection reading, with the stricter 8/10 printed in the same gate-table row**, both readings in the result file, the narrowing written into `docs/gates.md`, and tests that fail if the stricter number ever disappears. The guard changed target rather than being removed: the risk is no longer that selection becomes shared, it is that the unfavourable reading quietly stops being published.

The provenance is stated plainly in the new file instead of claimed away: normals from the depth grid and depth-edge rejection are `cozmo-scan's` approach. Central differences on a depth grid have few degrees of freedom and I had their implementation open, so the file says so rather than asserting an independence it cannot demonstrate.

3. Public defect data for A-DMG-DETECT — DECLINED

What it was. A-DMG-DETECT reads NOT BUILT. `cozmo-scan` validated damage *classes* on 793 real defect photographs. The candidate staged synthetic protrusions at 14/20/25/30/50 mm — sizes derived from our detector's own declared thresholds rather than from anything on the truth side — and moved the gate from NOT MEASURED to measured-and-failing.

Why it is declined. The direction is honest: every published number got worse, and measured-and-failing beats unmeasured. But one reviewer found a verbatim copied block in the new benchmark, two found choices flowing from the truth side, and one could not confirm the numbers came from a real run. Three flags on a gate that ends up failing anyway is not a trade worth making this close to a deadline.

The honest position. Our detector is *geometric* — it finds departures from the wall plane and infers class from shape, reporting `class_source: "shape"` with confidence ≤ 0.5 . An appearance dataset of photographs cannot validate that; it would validate an appearance classifier we do not have. Building one to use the dataset would be adding a capability and calling it a validation. A-DMG-DETECT stays NOT BUILT, which is what it is.

4. HouseLayout3D for adjacency — DECLINED by its own author, with a finding worth keeping

What it was. Our `plan.adjacency` has never been checked against anything — the compliance matrix says so at 2.2. `cozmo-scan` validates its stitch solver on HouseLayout3D. The candidate measured our adjacency against it across 27 storeys of 16 real buildings, 225 truth rooms and 139 truth door pairs.

What it measured. Badly. Exact adjacency 1/27, 0/27, 3/27 and 3/27 across four input regimes; precision 0.23–0.32, recall 0.07–0.34.

Why it is declined. The implementing investigation recommended against adopting it, and the reason is sound: **the dataset ships buildings, not captures.** It provides layout meshes, door corner lists and

nerfstudio poses with no images and no depth behind them. There is nothing our pipeline can ingest, so the comparison requires synthesising a capture from the mesh — and then the result measures the synthesiser as much as the pipeline. The low scores above are partly an artifact of that, which makes them unusable as a validation of our adjacency.

The finding worth keeping. The same investigation produced a diagnostic that explains something this project has been documenting as a failure for days. Across those buildings there are 240 truth rooms, but **the layout geometry alone separates only 119 floor regions — half the truth's rooms are not separated by any geometric boundary at all.**

Our room splitting works on floor coverage: it divides space at doorway-width necks. If half of real rooms are not geometrically separated from their neighbours, then no amount of tuning a global erosion width recovers them, because the information is not in the floor plan. That is consistent with what [docs/fix_loop.md](#) concluded after two failed attempts — that room splitting here is not a parameter problem — and it is the first external evidence for *why*.

It also bounds the opponent's advantage honestly: they report 3/8/9 rooms where we report 2/5/5, and their wall-run approach reads structure our coverage approach cannot. But a method that works from floor coverage has a ceiling imposed by the data, and this is roughly where it sits.

(This diagnostic was measured by the investigation and is reported as its finding; unlike the numbers in the shipped commits, I have not independently re-derived it.)

What this page is for

The brief says confident garbage caps the total score. The cheapest way to produce confident garbage is to adopt a change because it moved a number, and the cheapest way to avoid it is to measure candidates in isolation and let someone else attack the result before it ships.

Two of six survived that. The four here would each have improved a headline number, and three of the four would have done it by narrowing a question, copying an implementation while claiming otherwise, or choosing a constant for its effect on the gates. The fourth was declined by the person who built it.